Volvo Trucks - CH3 SVC

Camera Interface Specification

Version 0.2 **DRAFT**

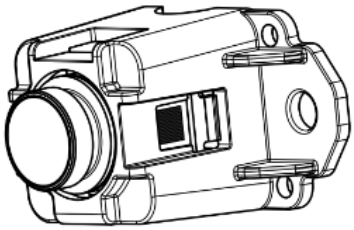
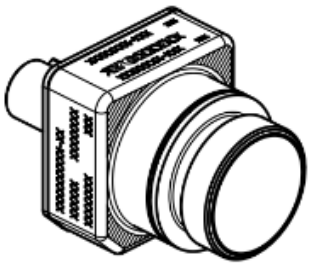
Document Number: 0220DN00KW

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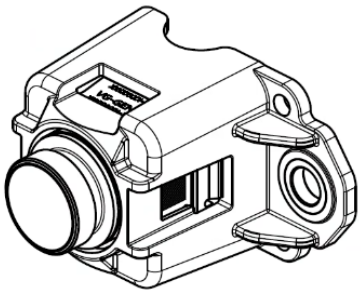
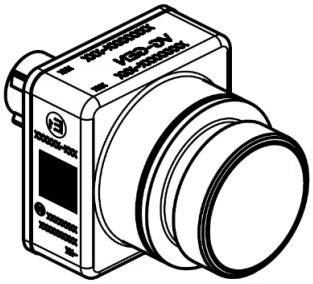
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Name of Customer:	Volvo Trucks	
Bosch Product Name*	CH3 SVC incl. Bracket	CH3 SVC
Volvo Trucks Product Name*	Near Range Camera incl. Bracket	Near Range Camera
Mounting location	1. Front apron 2. Side mirror / side wing	Rear view shutter module
Connector	FAKRA	FAKRA
B1-sample		
Customer Part Number	24569898-B02	24470537-B02
Customer Part Name	B1-sample	B1-sample
Bosch Part Number	0275B02079-01	0275B02065-01
Offer drawing number	0275A01208	0275A01208
Product Picture	-	-
Drawing Version	-	-
Drawing Date	-	-
B2-sample		
Customer Part Number	24569898.B02	24918084.B01
Customer Part Name	B2-sample	B2-sample
Bosch Part Number	0275B02093-01	0275B02094-01
Offer drawing number	0273A02227	0273A02234
Product Picture		
Drawing Version	01	00
Drawing Date	2025-02-07	2025-03-19

*: from section 1 onwards, component is just called camera

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C1-sample		
Customer Part Number	24569898.B03	24918084.C02
Customer Part Name	C1-sample	C1-sample
Bosch Part Number	0275B02127-01	0275B02128-01
Offer drawing number	0273A02229	0273A02239
Product Picture		
Drawing Version	06	01
Drawing Date	2025-09-15	2025-07-03

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1. Scope

The purpose of this document is to inform about the Camera Interface in detail.

This document is aimed to be shared with Volvo Trucks. Sending the document to other third-party component suppliers is not allowed. Exceptions need to be agreed by Robert Bosch GmbH.

This Camera Interface Specification is valid for the following CH3 SVC samples:

B1-sample		
Component	Manufacturer	Version Major
Image Sensor	Samsung	S5K3B6SAA3-Z072 KY2KAV
Serializer	Analog Devices / Maxim	MAX96717FGTJ/VY+T
PMIC	Rohm	BD868F6MUF-CE2
EEPROM	Rohm	BR24H32
I2C-Settings		
Serializer	SER-Setting-Name: “SVC3B6_SerDes_VolvoTruck_SXPF_I2C_Pixel_20250211.dvreg”	
Image Sensor	• IMG-Setting-Name (application relevant HDR setting): “S5K3B6SA_MAXIM_HDR4_RSSR_12B_ MCLK25_60fps_1920x1540_MIPI4_620Mbps_ wGamma_wEMBD_20250204.sset” • Originator of setting: Samsung • Embedded data: ON	
Functional Safety Features (relates to complete Camera)		
Image Sensor	• Defined via I2C-setting. • Features: Samsung default configuration • I2C CRC: OFF	
PMIC	• Failure reactions: According to platform concept • I2C CRC: OFF	
Cybersecurity Features (relates to Image Sensor only)		
Key fusing	NO	
Device authentivication	OFF	
Image authentification	OFF	
EEPROM authenticity and integrity	OFF	

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B2-sample		
Component	Manufacturer	Version Major
Image Sensor	Samsung	S5K3B6SAA3-Z172-KA2KAV
Serializer	Analog Devices / Maxim	MAX96717FGTJ/VY+T
PMIC	Rohm	BD868F6MUF-CE2
EEPROM	Rohm	BR24H32
I2C-Settings		
Serializer	SER-Setting-Name: “SVC3B6_SerDes_VolvoTruck_SXPF_I2C_Pixel_20250211.dvreg”	
Image Sensor	• IMG-Setting-Name (application relevant HDR setting): “S5K3B6SA_MAXIM_HDR4_RSSR_12B_ MCLK25_60fps_1920x1540_MIPI4_620Mbps_ wGamma_wEMBD_20250204.sset” • Originator of setting: Samsung • Embedded data: ON	
Functional Safety Features (relates to complete Camera)		
Image Sensor	• Defined via I2C-setting. • Features: Samsung default configuration • I2C CRC: OFF	
PMIC	• Failure reactions: According to platform concept • I2C CRC: OFF	
Cybersecurity Features (relates to Image Sensor only)		
Key fusing	NO	
Device authentivication	OFF	
Image authentification	OFF	
EEPROM authenticity and integrity	OFF	

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C1-sample and C2-sample		
Component	Manufacturer	Version Major
Image Sensor	Samsung	S5K3B6SAA3-Z172-KA2KAV
Serializer	Analog Devices / Maxim	MAX96717FGTJ/VY+T
PMIC	Rohm	BD868W0MUH-C
EEPROM	Rohm	BR24H32
I2C-Settings		
Serializer	SER-Setting-Name: “SVC3B6_SerDes_VolvoTruck_SXPF_I2C_Pixel_C1_20250828”	
Image Sensor	**Setting compatibility:** **B2-sample** settings: • IMG-Setting-Name (application relevant HDR setting): “S5K3B6SA_MAXIM_HDR4_RSSR_12B_ MCLK25_60fps_1920x1540_MIPI4_620Mbps_ wGamma_wEMBD_20250204.sset” • Originator of setting: Samsung • Embedded data: OFF **C-sample** settings: target settings, not released yet • Originator of setting: Samsung • Embedded data: ON	
Functional Safety Features (relates to complete Camera)		
Image Sensor	With **B2-sample** settings: • Defined via I2C-setting. • Features: Samsung default configuration • I2C CRC: OFF With **C-sample** Settings: • Defined via I2C-setting. • Features: According to Volvo Truck concept • I2C CRC: ON	
PMIC	• Failure reactions: According to Volvo Truck concept • I2C CRC: On	
Cybersecurity Features (relates to Image Sensor only)		
Key fusing	Yes	
Device authentivication	Available	
Image authentification	Available	
EEPROM authenticity and integrity	Available	

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2. I2C addresses

Component	I2C Address [8-Bit]
Image Sensor	0x20
Serializer	0x80
PMIC	0xE0
EEPROM	0xA0

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3. MIPI D-PHY Routing

Image Sensor Output		Serializer Input	
Lane	Polarity	Lane	Polarity
CLK	N	CLK	P
	P		N
0	N	1	P
	P		N
1	N	0	P
	P		N
2	N	2	P
	P		N
3	N	3	N
	P		P

Register setting of Serializer to accommodate lane and polarity swaps [Slave Address (8-Bit), Register Address (16-Bit), Register Value (8-Bit)]:

- 0x80, 0x0332, 0xE0
- 0x80, 0x0333, 0x01
- 0x80, 0x0334, 0x10
- 0x80, 0x0335, 0x07

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4. Serializer Configuration

B1-sample and B2-sample					
MFP[1]	**Name of net**	**Description of net**	**External Pulls**	**Default state (camera powered, no config of camera)**	**Target Configuration of MFP**
0	IMG_RST_N	Enable (high) or disable (low) imager	Pull down	Low	Push-pull high or Low
1	CFG0_FSYNC	Strap during start-up; Frame Sync connection to imager	Strap set to be I2C, XTAL, 0x80		Push-pull high or Low
2	CFG1	Strap during start-up;	Strap set to be COAX, Pixel, 3Gbps		
3	SER_LH_EN	Enable (high) or disable (low) lens heating	Pull down	Low	Push-pull high or Low
4	SER_CLKOUT	Programmable clock output of the serializer; Used as imager clock input		Low	25MHz
5	PMIC_ERROR_N	Error input from PMIC; No error (high), error (low)		High or low	Input internal pull down
6	VARIANT_CODE	Voltage divider input: low: B1-Sample (w/ LH)	Pull down	Low	Input No internal pull
7	IMG_ERROR_N	Error input from imager; No error (high), error (low)	Pull down	High or low	Input internal pull down
8	EEPROM_WP	Write protect of EEPROM: high: protected low: write enabled	Pull up	High	Push-pull high or Low

[1] MFP: Multi-Functional PIN

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C1-sample and C2-sample					
MFP[2]	**Name of net**	**Description of net**	**External Pulls**	**Default state (camera powered, no config of camera)**	**Target Configuration of MFP**
0	IMG_RST_N	Enable (high) or disable (low) imager	Pull down	Low	Push-pull high or Low
1	CFG0_FSYNC	Strap during start-up; Frame Sync connection to imager	Strap set to be I2C, XTAL, 0x80		Push-pull high or Low
2	CFG1	Strap during start-up;	Strap set to be COAX, Pixel, 3Gbps		
3	SER_LH_EN	Enable (high) or disable (low) lens heating	Pull down	Low	Push-pull high or Low
4	SER_CLKOUT	Programmable clock output of the serializer; Used as imager clock input		Low	25MHz
5	PMIC_ERROR_N	Error input from PMIC; No error (high), error (low)		High or low	Input internal pull down
6	LH_Readback	Digital readback of lens heating voltage: lens heating enabled (high) lens heating disabled (low)	Voltage divider to meet digital IO levels	Low	Input No internal pull
7	IMG_ERROR_N	Error input from imager; No error (high), error (low)	Pull down	High or low	Input internal pull down
8	EEPROM_WP	Write protect of EEPROM: high: protected low: write enabled	Pull up	High	Push-pull high or Low

[2] MFP: Multi-Functional PIN

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5. EEPROM

5.1. EEPROM General

This document refers to this Bosch EEPROM layout (refer also to description of general header block):

- EEPROM Layout version = Version 1.0, 2024-07-24
- EEPROM Customer = 0x05
- EEPROM Customer Variant Flag = 0x00
- VersionMajor = 0x00
- VersionMinor = 0x00

EEPROM layout history:

EEPROM layout version	Version Major	Version Minor	Changes
Version 1.0	0x00	0x00	Baseline

EEPROM layout camera history:

Sample	Version Major	Version Minor	Remark
CH3 SVC B1-sample	0x00	0x00	
CH3 SVC B2-sample	0x00	0x00	
CH3 SVC C1-sample	0x00	0x00	

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5.2. EEPROM Memory

This section describes EEPROM memory of camera.

Figure 1 below describes block diagram of CH3 SVC camera excluding system integration.

Camera contains EEPROM memory.

Size of EEPROM memory is 32 kBit.

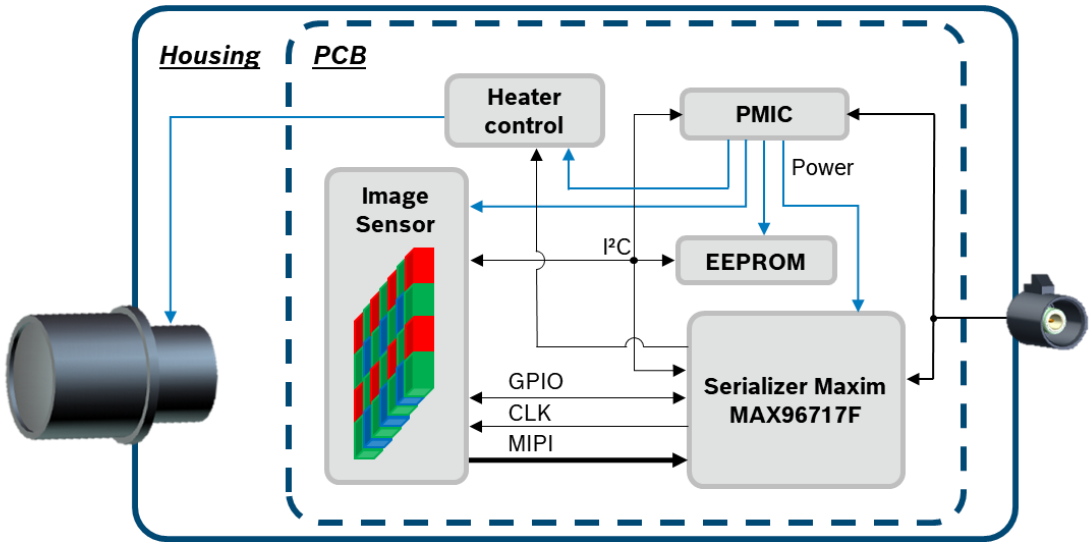


Figure 1: Camera-level block diagram

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5.3. EEPROM MAP

5.3.1. Layout

Figure 2 below describes basic overview of EEPROM layout. All blocks are mentioned including size and planned access. Read-Only in column *Access by ECU* does not mean that write protection is implemented but does mean that this block shall not be accessed by ECU. Column *Write access* describes who shall do write access to this block.

No.	Block Name	Size w/ optional data	Size w/o optional data	Operatio-nal data	Access by ECU	Write access
1	General Header Block	32	32	yes	Read-Only	Bosch plant
2	Label Data Block	224	224	yes	Read-Only	Bosch plant
3	Intrinsic Data Block	352	352	yes	Read-Only	Bosch plant
4	Block for Bosch plant data	64	64	yes	Read-Only	Bosch plant
5	Security (RO Data) Block	96	96	yes	Read-Only	Bosch plant
6	OEM Data Block	64	64	no	Read-Write	ECU
7	Optional: Failure Entry block	448		no	Read-Write	ECU
8	Optional: Temperature Histogr. Block	96		no	Read-Write	ECU
	used space [Bytes]	1376	832			
	free space [Bytes]	2719	3263			
	degree of filling	33.6%	20.3%			

Figure 2: EEPROM layout overiew (1)

Figure 3 below describes more details of the blocks of the EEPROM layout. ID, name, start address, size, basic contents and use of signature and security per block are described.

#ID	Block Name	Start-address	Size	Block description	Signature (Security) / CRC (Safety)
0x01	General Header Block	0x0000	32	General information about the EEPROM	CRC for each data block
0x02	Label Data Block	0x0020	224	Label data of the camera for traceability	
0x04	Intrinsic Data Block F-Theta model	0x0100	352	Determined parameters during intrinsic	
0x05	Internal Part Information Block	0x0260	64	Placeholder for internal part information	
0x0F	Security (Bosch RO Data) Block	0x02A0	96	Contains security signature for all read only blocks above written in	ECDSA-Signatur for RO data (RO blocks above) & CRC.
0x10	OEM Data Block	0x0300	64	Contains specific identifier data written by OEM	CMAC-Signature and CRC for each data block CRC is not part of CMAC calculation, but CRC includes CMAC value (CRC secures signature)
0x11	Failure Entries Block	0x0340	832	Failure entries defined by Bosch & customer	
0x12	Temperature Histogram Block	0x0680	96	Histogram of operating temperature of	

Figure 3: EEPROM layout overiew (2)

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5.3.2. Data storage

Data stored in "Big-Endian" Format.

Data format of intrinsic data is "32-bit single precision" or "64-bit double precision" according to IEEE754 (refer to description of parameters for detailed information whether parameter is single or double precision).

5.3.3. Block structure

EEPROM layout is organized in blocks. Each block has a certain size and CRC checksum is calculated and stored per block.

Data in blocks consists of several elements. Each element contains some data.

Address offset of element is relative to start address of block.

Size represents size in bytes of data type. Field counter describes how many fields of data type are stored in the lement. Length describes length of element.

Example: An element might contain 6 fields of unit16. Size is 2 bytes because unit16 is 2 bytes. Field counter is 6 because element contains 6 fields. Length of element is 2 x 6 = 12 bytes.

This is data stored in first three elements of each block.

Address offset	Data type	Size [byte]	Field counter	Length [byte]	Parameter Name	Remark
0	uint8	1	1	1	block type ID	
1	unit16	2	1	2	block length	Total number of bytes used for block
3	unit24	3	1	3	block CRC write counter	Incremented at each write of CRC

This is data stored in last element of each block.

Address offset	Data type	Size [byte]	Field counter	Length [byte]	Parameter Name	Remark
Block length -2	uint16	2	1	2	CRC checksum	Checksum calculated over all fields of the Block except crc

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5.3.4. Calculation of CRC checksum

This is CRC Algo for individual data blocks: CRC-16/MCRF4XX.

- Polynomial: 0x1021
- Initial value for CRC calculation: 0xFFFF
- Final Xor Value (XorOut): 0x0000
- Input Reflected (RefIn) / Output Reflected (RefOut) = true
- Example: String "Hello World!"
    - Converted to Hex Format: 0x48 0x65 0x6C 0x6C 0x6F 0x20 0x57 0x6F 0x72 0x6C 0x64 0x21 -> Resulting CRC: 0xF444

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5.3.5. General header block

Table below lists the information stored in the general header block. Length of general header block is 32.

Address offset [dec]	Data type	Size [byte]	Field counter	Length [byte]	Parameter Name	Remark
0	uint8	1	1	1	block type ID	2
1	unit16	2	1	2	block length	224
3	unit24	3	1	3	block CRC write counter	
6	unit32	4	1	4	...	
10	uint8	1	1	1	EEPROM Customer Flag	Volvo Trucks = 0x05
11	uint8	1	1	1	EEPROM Customer Variant Flag	SVC = 0x00 (F-Theta intrinsic block)
12	uint8	1	1	1	Major version	0x00
13	uint8	1	1	1	Minor version	0x00
14	uint8	1	1	1	...	
15	unit8	1	15	15	...	
30	uint16	2	1	2	CRC checksum	

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5.3.6. Label data block

Table below lists the information about the label information stored in the label data block. Length of label data block is 224.

Address offset [dec]	Data type	Size [byte]	Field counter	Length [byte]	Parameter Name	Remark
0	uint8	1	1	1	block type ID	2
1	unit16	2	1	2	block length	224
3	unit24	3	1	3	block CRC write counter	
…						
8	unit8	1	16	16	Imager-ID	
24	unit8	1	1	1	Imager Revision	
25	string	1	10	10	Bosch part number	written on label
…						
45	string	1	3	3	Index of Bosch P/N	written on label
…						
64	string	1	3	3	Bosch plant code	Blaichach is “050”
67	string	1	8	8	Production date “yyyymmdd”	Part of unique serial number
…						
83	string	1	4	4	Line on which camera was produced	Part of unique serial number
87	unit32	4	1	4	counts cameras pro-duced per day	Part of unique serial number
…						
111	string	1	24	24	Lens DMC contents	
…						

See continuation on next page.

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Continuation:

Address offset [dec]	Data type	Size [byte]	Field counter	Length [byte]	Parameter Name	Remark
135	string	1	6	6	ID of OEM	World manufacturing code
141	string	1	20	20	OEM part number (20 digits)	Written on label. • Bracket variant: bracket assembly part number • CAM-only variant: camera part number
170	string	1	30	30	OEM Serial No	Currently not used.
...						
222	uint16	2	1	2	CRC checksum	

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5.3.7. Intrinsic data block (F-Theta)

Table below lists the information about the intrinsic parameter stored in the intrinsic data block. Length of label data block is 352. Model is F-Theta.

Address offset [dec]	Data type	Size [byte]	Field counter	Length [byte]	Parameter Name	Remark
0	uint8	1	1	1	block type ID	4
1	unit16	2	1	2	block length	352
3	unit24	3	1	3	block CRC write counter	
...						
8	unit16	2	2	4	Calibration Model Version	0x0001, 0x0001 (F Theta 1.1)
12	unit8	1	1	1	Intrinsic Calibration Model	0x0A (F Theta)
13	string	1	8	8	Bosch software version	
21	uint16	2	1	2	Image rows	Number of rows of image the intrinsic is referenced to (full sensor read-out ROI)
23	uint16	2	1	2	Image columns	Number of columns of image the intrinsic is referenced to (full sensor read-out ROI)
25	uint16	2	1	2	Start Row	Start Row of ROI, End Row = Imager Rows - Start Row
27	uint16	2	1	2	Start column	Start Column of ROI, End Column = Image Columns - Start Column
29	float32	4	1	4	Camera constant along y-direction	[pix]
33	float32	4	1	4	Aspect ratio	
37	float32	4	1	4	Principal point x	[pix]
41	float32	4	1	4	Principal point y	[pix]
45	float32	4	12	48	Distortion parameters for transformation from undistorted to distorted image (U2D)	= [A1, A2, A3, A4, A5, A6, A7, A8, B1, B2, C2, r0]
93	float32	4	12	48	Undistortion parameters for transformation from distorted to undistorted image (D2U)	= [A1, A2, A3, A4, A5, A6, A7, A8, B1, B2, C2, r0]

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...						
141	Float32	4	3	12	Internal Angles	Orientation of the camera wrt. Nominal orientation in the fixture coordinate-system in [°] defined in DIN70k-coord. Sys. = [Roll, Pitch, Yaw].
...						
231	float32	4	1	4	Valid FOV	Field of View within which the intrinsic is valid in [°]
235	float32	4	1	4	Valid Image Radius	Image-Radius within which the intrinsic is valid in [pix]
...						
350	uint16	2	1	2	CRC checksum	

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6. Lens Heating Control

6.1. Disclaimer

- Please refer to the conditions provided TCD[3] (Document Number: 0220DN00E0).
- Information to be used as reference only. Depends on specific use-case.

6.2. Activation of lens heating

Please refer to:

- MFP controls in Serializer Configuration in section 4.
- Description contained in the TCD.

[3] TCD: Technical Customer Documentation

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7. Temperature Sensors

The camera features multiple temperature sensors:

Device	Description	Temperature Threshold	Accuracy
PMIC	Thermal Warning	140 °C	+/- 10 °C
	Thermal Shutdown	175 °C	-25 °C to +10 °C
Lens Heating Switch	Thermal Shutdown	160 °C (30 °C Hysteresis)	n/a
Serializer	Temperature Sensor 1	Programmable	-40 °C – 105 °C: +/- 3.0 K 105 °C – 125 °C: +/-2.5 K
	Temperature Sensor 2		
Image Sensor	Temperature Sensor 1	Programmable	-40 °C – 105 °C: +/- 3.0 K 105 °C – 115 °C: +/- 2.0 K 115 °C – 125 °C: +/- 1.0 K
	Temperature Sensor 2		

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8. Image Sensor Settings

8.1. General

The following summary provides an overview of the Samsung 3B6 image sensor setting which was used in Design Validation.
During product development this setting might experience changes.

8.2. Function Overview

B1-sample and B2-sample		
Function	**Setting 1**	**Setting 2**
Short Description	**60 fps** considered **DV-setting**	**30 fps** setting
General Parameters		
Resolution	1920 px x 1536 px	←
Bit depth per Pixel	12bit	16 bit
Frame rate	60 fps	30 fps
INCK	25 MHz	←
Sync Mode	Optional. Ext. trigger (accuracy tolerances need to be kept)	←
Gains	Use recommended gains from supplier.	←
Exposure time	11 ms (adaptable)	←
MIPI non-continuous mode	ON	←
Frame Format		
Image orientation	No Flip, No Mirror	←
Optical center	No decentering	←
Embedded data (Frame Format)	Activated, fixed	←
Data type of embedded data	same type as image (0x2C)	←
Statistics	OFF	←
Frame Sequence	No context switching, VC0	←
HDR Frame		
Frame Mode	HDR4	←
Apply Factor	OFF	←
Piecewise linearization	Recommended setting from supplier used.	←
HDR merger configuration	Use recommended setting from supplier.	←
Phase correction	Use recommended setting from supplier.	←
Post Processing		
Bad Pixel Correction	OFF	←
Noise Reduction	OFF	←
White Balance	OFF	←
Auto Exposure	OFF	←
Functional Safety		
Imager features	VT-specific configuration	←

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8.3. Error Flags Configuration

- Error pin is set to “active low” i.e. low in case of an error.
- Error flags value is per default “1” and it will switch to “0” in case of an error. (valid for error bits in registers and embedded data)

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9. Revison

This section describes revision of the document.

Revision	Date	Author	Status	Description
0.1	2025-06-03	Marc Schmid	Draft	• Initial revision, incomplete
0.2	2025-10-13	Marc Schmid, Annika Wilms	Draft	• Document updated for C1-sample

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